RIT OPPORTUNITY DATA

From Raw Records to Predictive Insights

A Full, Detailed Consolidated Report on Student Participation, Engagement, and Success Prediction

Weeks 1 – 4 · Data Understanding · Cleaning · Cohort Analysis · EDA · Predictive Modeling · Recommendations

Rameen Shahzad
Lorenzo Miguel
Iqra Muqaddas
Ronan Patrick
Johnson Modinat
Vaishnavi Kedare
Yvvone Muli
Anushka Poshattiwar

Excelerate Externship Program · 2026

Final Report and Presentation Video:
https://drive.google.com/drive/folders/1430pffXZnNGsJG812uSyaIsCTCcl1Fcr?usp=drive_link

Project Roadmap

Four weeks, one continuous analytical thread — from raw CSV to deployable prediction model and a strategic action plan

01

Week 1

Dataset Understanding & Initial Cleaning

Data dictionary, quality assessment, first-pass cleaning of text, name, and date fields

02

Week 2

Cohort Analysis & Exploratory Data Analysis

Feature engineering, cohort segmentation, demographic, temporal, and interpretive analysis

03

Week 3

Predictive Modeling

Five ML models trained, compared, and interpreted to predict participation success

04

Week 4

Insights & Recommendations

Evidence-based strategy, KPIs, and process reflections for future cohorts

WEEK 1

Understanding the Dataset

Student participation records across courses, internships, competitions, events, and engagement programs

8,558

cleaned records

16

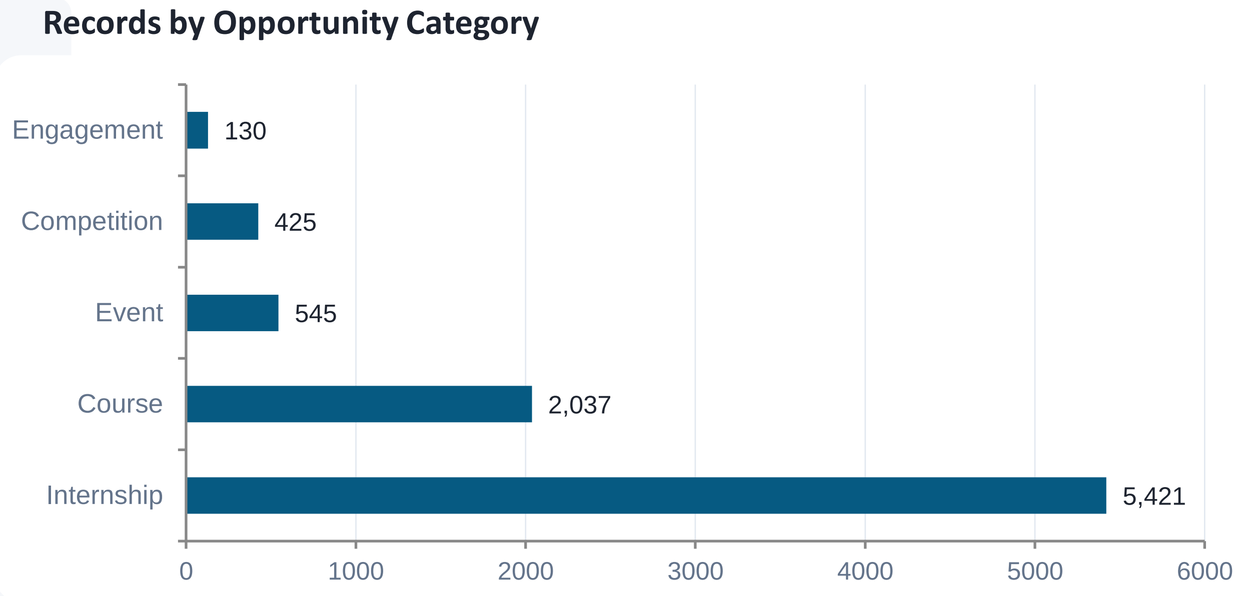
variables

30+

countries represented

5

opportunity categories



Five opportunity types — Courses, Internships, Competitions, Events, and Engagement Programs — span the full student journey from sign-up through application to final status resolution, drawing on learners from 30+ countries.

WEEK 1

Data Dictionary

Core variables used across all four weeks of analysis

Variable	Description	Type	Example
Learner SignUp DateTime	When the student registered on the platform	DateTime	06/14/2023 12:30
Opportunity Category	Course, Internship, Competition, Event, Engagement	Categorical	Internship
Apply Date	When the student submitted their application	DateTime	06/14/2023 12:36
Status Code / Description	Numeric / text code tracking outcome progression	Categorical	1070 / Team Allocated
Country	Country of residence or nationality	String	Pakistan, India, US
Current/Intended Major	Student's field of study	String	Computer Science
Date of Birth	Used to derive Age for eligibility & segmentation	Date	01/12/2001
Opportunity Start/End Date	Temporal bounds of the opportunity	DateTime	11/03/2022 – 06/29/2024

Status Code system: 1010 Applied · 1030 Rejected · 1040 Waitlisted · 1050 Dropped Out · 1070 Team Allocated · 1080 Started · 1110 Withdraw · 1120 Rewards Award

WEEK 1

Dataset Quality Assessment

Five categories of quality issues were identified prior to preprocessing

Quality Check	Observation	Potential Impact
Missing Values	Some records missing dates and status info	Affects timeline & progression analysis
Invalid Date Formats	Malformed timestamps, e.g. 1085640:21:29	Requires correction during preprocessing
Test Records	Sample/placeholder entries present	Must be removed to avoid skewing results
Data Consistency	Variations in country, institution, major names	Standardization needed for grouping
Duplicate Entries	Potential duplicate learner records	Can inflate participation statistics

As the participation funnel shows, the most significant drop-offs occur at the application review stage (rejection / waitlisting) and again after team allocation (dropout / withdrawal) — precisely the stages most susceptible to missing or inconsistent status records, reinforcing the need for thorough cleaning before analysis.

WEEK 1

Text Field Cleaning

Institution names, learner names, and academic majors were standardized field-by-field

6

Institution Names

Leading/trailing spaces trimmed, converted to Title Case; non-alphabetic entries (e.g. "123456") set to null.

"saint louis university" → "Saint Louis University"

29

Learner First Names

Removed extra spaces, standardized to Title Case; entries with numbers/symbols or placeholder "Nan" nulled out.

Ensures identity fields contain only meaningful values

110

Academic Majors

Placeholder responses like "Oth", "Fresher", "Student", "Na" consolidated into one category.

All variants → "Undeclared/Unknown"

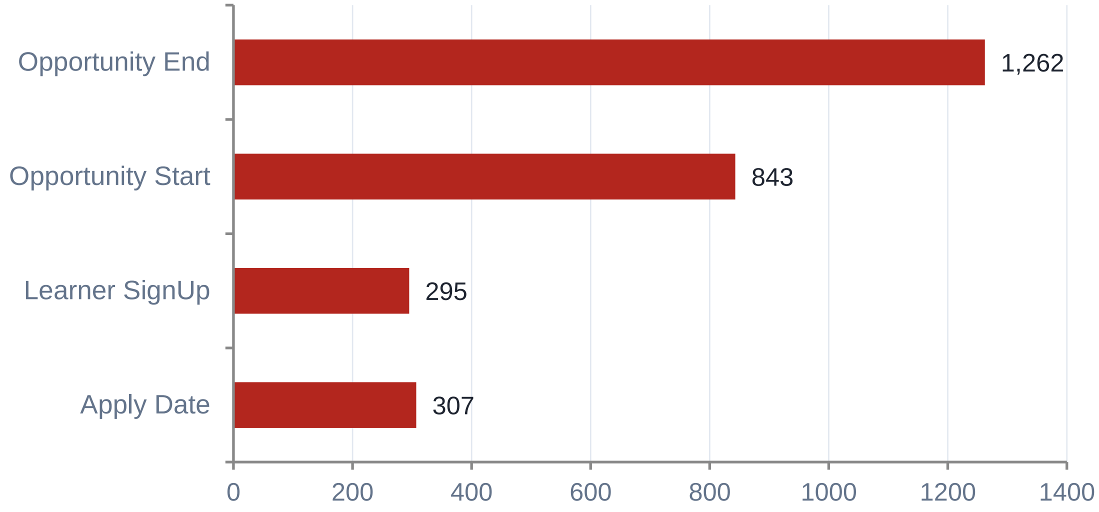
Records affected shown as callouts above · All corrections preserved the underlying record rather than deleting it

WEEK 1

Date Field Cleaning & Cumulative Summary

Malformed timestamps were coerced to NaT using pandas errors='coerce'

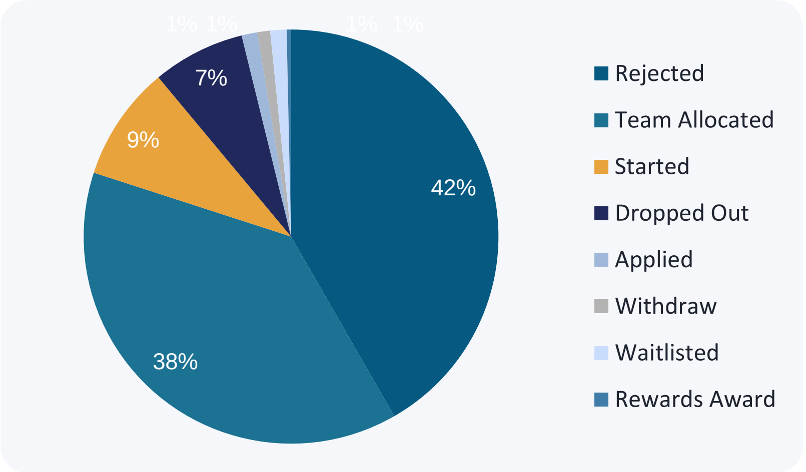
Corrupted Entries by Date Field



Summary of Week 1 Cleaning Activities

Activity	Records Affected
Test records flagged	5
Invalid first names cleaned	29
Invalid institution names cleaned	6
Placeholder majors standardized	110
Corrupted date values corrected	2,700+

Final Status Breakdown (after cleaning)



Outcome: the cleaned dataset contains 8,558 records across 16 variables — Internship (5,421), Course (2,037), Event (545), Competition (425), Engagement (130) — ready for cohort analysis and feature engineering in Week 2.

WEEK 2

Cleaning Re-Verified & Extended

The Week 1 baseline was re-verified before new features were engineered — one new issue surfaced

Cleaning Step	Records Affected	Method / Notes
Apply Date repair	307 repaired	Status digits embedded in time component; regex fallback used
Datetime standardization	8,558 rows	5 fields converted to pandas Timestamp for duration arithmetic
Institution standardization	19+ variants	Whitespace stripped, Title Case applied
Categorical validation	0 invalid values	Gender, Category, Status mapping confirmed
Apply-before-Signup flag	43 flagged	Retained for investigation, not removed
Age outlier flag	40 flagged	Age < 15 or > 60 relative to 2024
Drop incomplete records	10 rows dropped	Missing both Institution Name and Major — only deletion step

WEEK 2

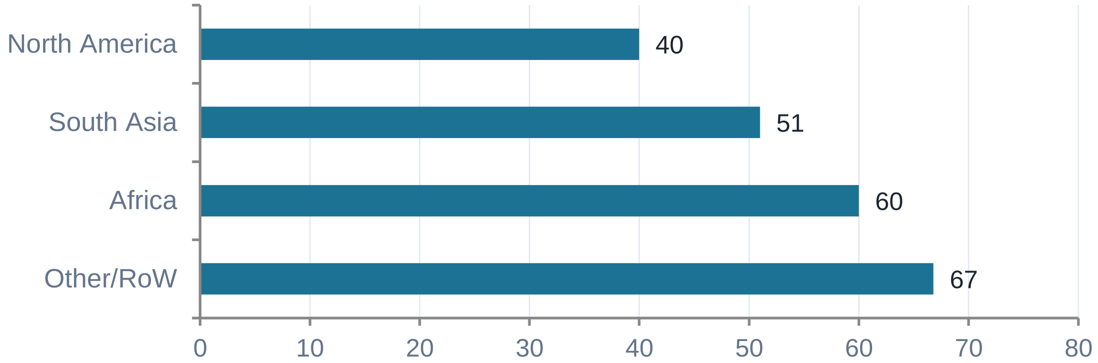
Cohort Analysis: Time & Region

Grouping learners by signup month and region reveals where the platform is winning — and losing — engagement

Monthly Sign-up Cohorts (Jan 2023 – Mar 2024)

Sign-up volume grew sharply through mid-2023 and again in Jan–Feb 2024, but the success-rate trendline moved the opposite direction — peaking near 85% in March 2023 and falling to roughly 25–33% by early 2024.

Success Rate by Region



Success & Dropout Rate by Region

Region	Learners	Success	Dropout
North America	3,979	40.0%	9.0%
South Asia	3,137	51.0%	8.2%
Africa	1,206	60.0%	5.5%
Other / RoW	226	66.8%	8.4%

Despite the largest learner base, North America has the lowest success rate and highest dropout of any region. Africa, a smaller cohort, shows the strongest outcomes — a gap worth investigating for selection criteria or opportunity-type differences.

WEEK 2

Opportunity Category Cohorts & Derived Features

Category is the starkest divide in the dataset; ten features were engineered to capture it

Category	Total	Success	Dropout	Dropped Out (n)
Internship	5,413	21.5%	12.4%	669
Course	2,035	93.2%	1.2%	24
Event	545	96.3%	0.4%	2
Competition	425	84.2%	1.2%	5
Engagement	130	96.9%	0.0%	0

Internships alone account for 669 of the dataset's ~700 total dropouts — effectively the entire attrition problem traces back to one opportunity type.

Age Profile: learner base is concentrated in the 20–27 age range, consistent with an undergraduate-to-early-career population — supporting Age as a continuous model feature.

10 Derived Features

- **Age**
  Demographic segmentation
- **Age Flag**
  Isolates data-entry errors
- **Signup Cohort**
  Month-based grouping
- **Time To Apply Days**
  Engagement speed proxy
- **Opportunity Duration Days**
  Short vs. long engagements
- **Has Start Date**
  Avoids biased imputation
- **Successful Outcome**
  Primary prediction target
- **Dropped Out Flag**
  Dedicated churn target
- **Learner Tenure Days**
  Control variable
- **Region**
  71 countries → 4 groups

WEEK 2

EDA — Participation Overview

8,588 applications from 3,842 students across 23 opportunities offered through 1,815 institutions

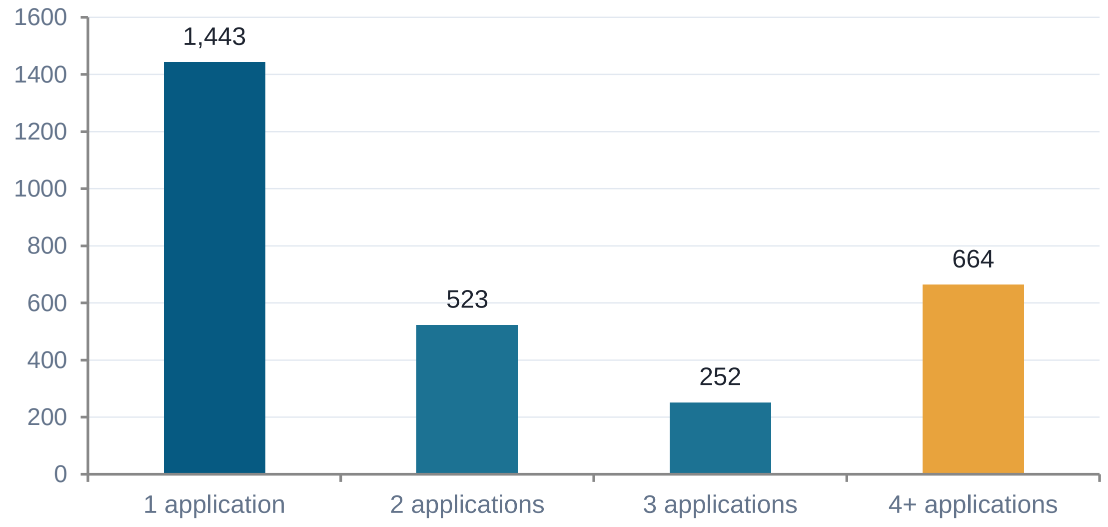
8,588
total records

3,842
unique students

23
unique opportunities

2.2
avg. applications / student

Applications per Student



What This Tells Us

1,443 students (largest group) submitted only one application. Yet 664 students applied to four or more opportunities : a sizable, highly-engaged segment that actively explores the platform. Internships accounted for over 60% of all applications, making them the most popular category, followed by courses. This indicates broad accessibility across institutions with internships driving most engagement.

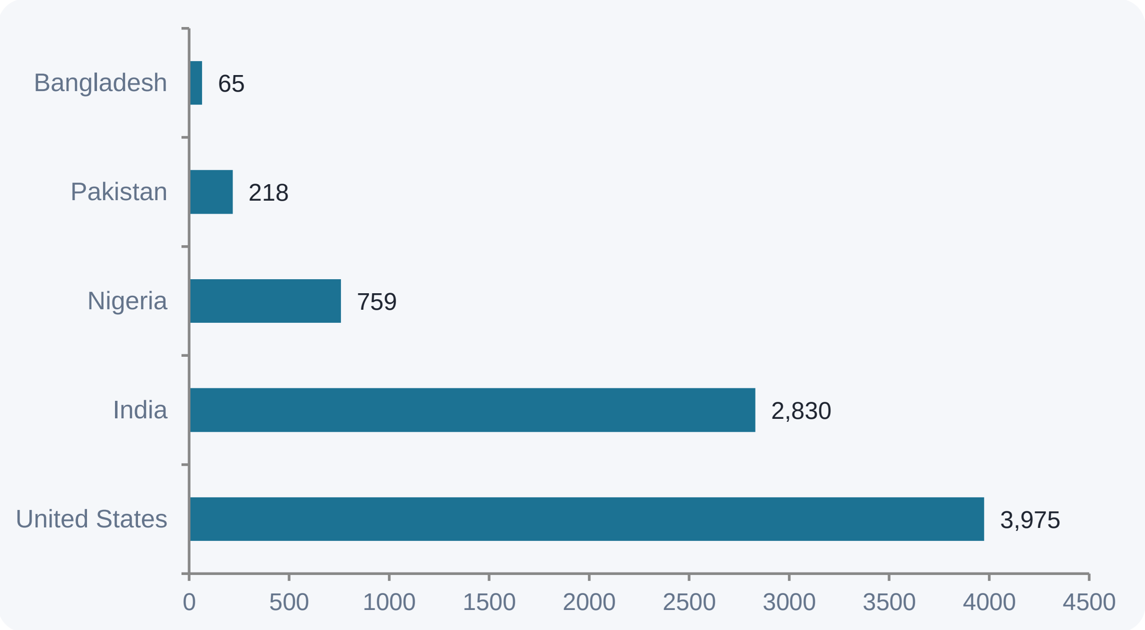
WEEK 2

Engagement Funnel & Status Breakdown

Rejected (41.7%) and Team Allocated (38.3%) dominate the outcome distribution

Code	Status	Count	% of Total
1010	Applied	105	1.2%
1030	Rejected	3,566	**41.7%**
1040	Waitlisted	109	1.3%
1050	Dropped Out	615	7.2%
1070	Team Allocated	3,273	**38.3%**
1080	Started	766	9.0%
1110	Withdraw	85	1.0%
1120	Rewards Award	29	0.3%

Top Countries by Applications

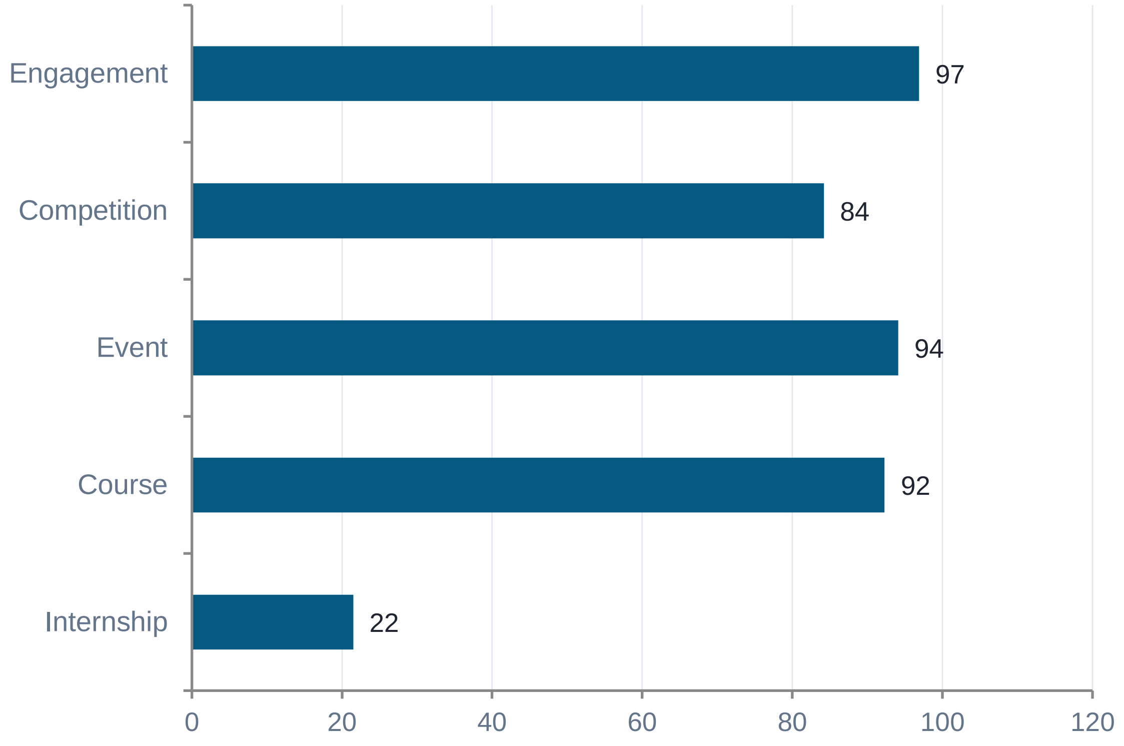


US and India together contribute ~80% of applications; eight of the top ten majors are technology-related, confirming a STEM-focused, career-development audience.

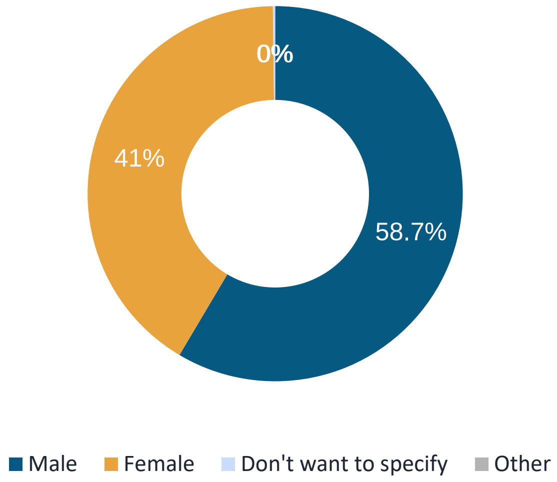
Acceptance Rate & Gender Distribution

Acceptance varies sharply by category; a moderate gender gap tracks STEM field composition

Acceptance Rate by Category



Gender Distribution



58.7%

male applicants

41%

female applicants

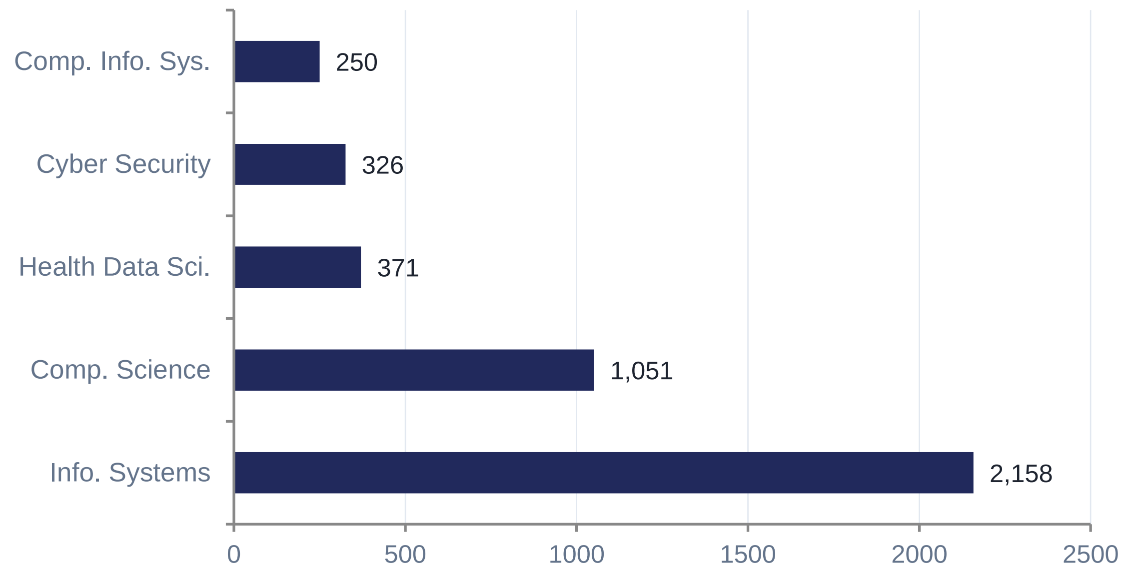
A gap of ~17.6 points, likely reflecting STEM field composition rather than platform bias.

WEEK 2

Fields of Study & Growth Over Time

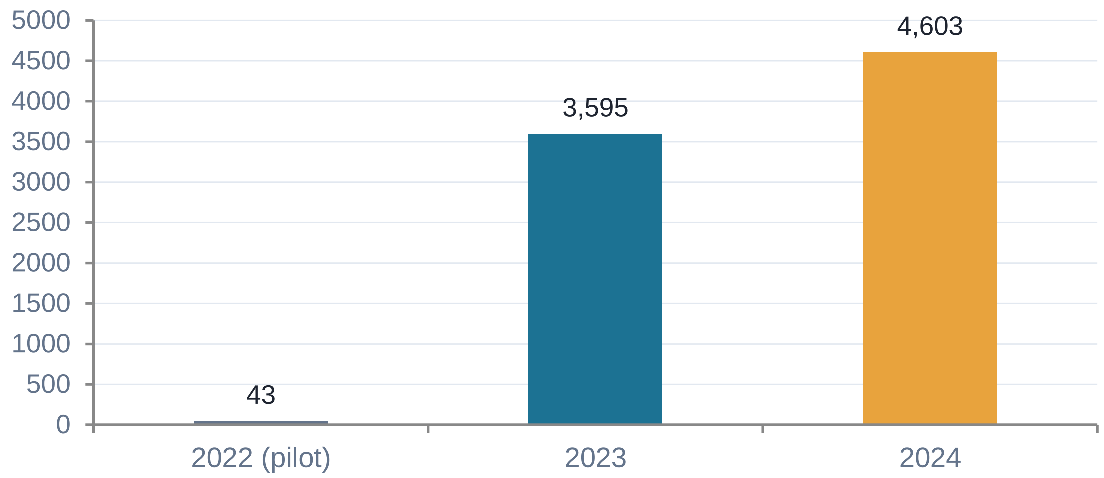
A STEM-dominant applicant pool experiencing consistent year-over-year growth

Top Fields of Study



Eight of the top ten majors are technology disciplines — confirming a STEM-focused, career-development audience.

Applications by Year



+28.0% growth from 2023 to 2024 — reflecting rising student adoption. If trends continue, annual applications could exceed 6,000 by 2025.

WEEK 2

Patterns, Anomalies & Key Findings

Seven EDA findings translate directly into platform recommendations

Internships dominate applications (63.3%)

but record the lowest acceptance rate (21.5%)

28% YoY application growth

strong platform trajectory, 2023 → 2024

US + India = ~80% of applications

a strong geographic concentration signal

54.2% missing Opportunity Start Date

limits duration-based and temporal analysis

8.2% post-acceptance attrition

615 dropped out, 85 withdrew after placement

43 records: Apply Date before Sign-up

likely batch-import or time-zone artifacts

17.6-point gender gap

58.7% male vs. 41.1% female, tracks STEM composition

WEEK 2

Interpretation — A Two-Tier Opportunity Landscape

Acceptance-rate data exposes a sharp structural divide across opportunity types

Category	Applications	Acceptance	Classification
Engagement	130	96.9%	Open access
Event	545	96.3%	Open access
Course	2,034	93.2%	Open access
Competition	425	84.2%	Moderately selective
Internship	5,413	**21.5%**	**Highly selective**

Because internships attract 63.3% of all applications, this creates a large-scale rejection experience for the majority of platform users — Opportunity Category is therefore likely to be the strongest predictor of success in any future model.12

Retention: Post-Acceptance Attrition

Status	% of Dataset
Dropped Out	7.2%
Withdraw	1.0%
Rejected (pre-placement)	41.6%
Team Allocated / Started	38.3%

A dropout after placement is more costly than a rejection at application — effort was already invested by both student and provider.

WEEK 2

Interpretation — Demographics, Growth & Data Quality

Concentration, gap, growth, and quality gaps each carry distinct implications

Finding	Interpretation	Implication
US + India = 79.6%	Reach is highly concentrated	Regional outreach & institutional partnerships
58.7% male applicants	Reflects STEM field composition	Monitor acceptance/dropout by gender
28% YoY growth	Platform scaling; supply must keep pace	Scale internship supply alongside demand
54.18% missing start dates	Duration analysis severely limited	Exclude from direct modeling; impute later
Category_Encoded corr. = 0.554	Opportunity type = strongest practical predictor	Prioritize as top feature in modeling

Data Quality Issues Carried Into Week 3

Opportunity Start Date

54.18% missing

Exclude as direct feature

Opportunity End Date

14.75% missing

Impute in Week 3

Apply Date

3.59% missing

Drop rows for time features

Preparing for Predictive Modeling

A binary classification problem: will a student reach a successful outcome?

8,547 total records

6,387 training records (80%)

1,597 testing records (20%)

47.6% baseline success rate

Target Variable

Target_Success

1 = Success: Started, Team Allocated, or Rewards Award

0 = Not Successful: Rejected, Dropped Out, Waitlisted, Applied, Withdraw

Stratified 80/20 split preserves class balance across training and test sets. Missing values checked column-by-column before modeling.

Missing Values Handled

Opportunity Start Date
54.18% *Exclude*

Opportunity End Date
14.75% *Impute*

Apply Date
3.59% *Drop rows*

Learner SignUp DateTime
3.45% *Drop rows*

Institution / Major
0.07% / 0.06% *Drop rows*

WEEK 3

Feature Engineering & Correlation with Target

Eight features engineered; correlation confirms Opportunity Category as the top practical predictor

Feature	Correlation	Interpretation
Status Code	0.881	Strongest — directly encodes outcome progression
Category_Encoded	0.554	Strong — opportunity type heavily determines success
Signup_Month	0.073	Weak positive — mild seasonal effect
Gender_Encoded	0.026	Very weak — minimal linear effect
Signup_Year	0.009	Negligible — cohort year alone not predictive
Days_Signup_to_Apply	-0.016	Negligible linear — may have non-linear value
Age	-0.019	Negligible — age alone not predictive
Apps_Per_Student	-0.086	Slight negative — high-volume applicants less targeted

Success Rate by Category

Engagement	96.9%
Event	96.3%
Course	93.2%
Competition	84.2%
Internship	21.5%

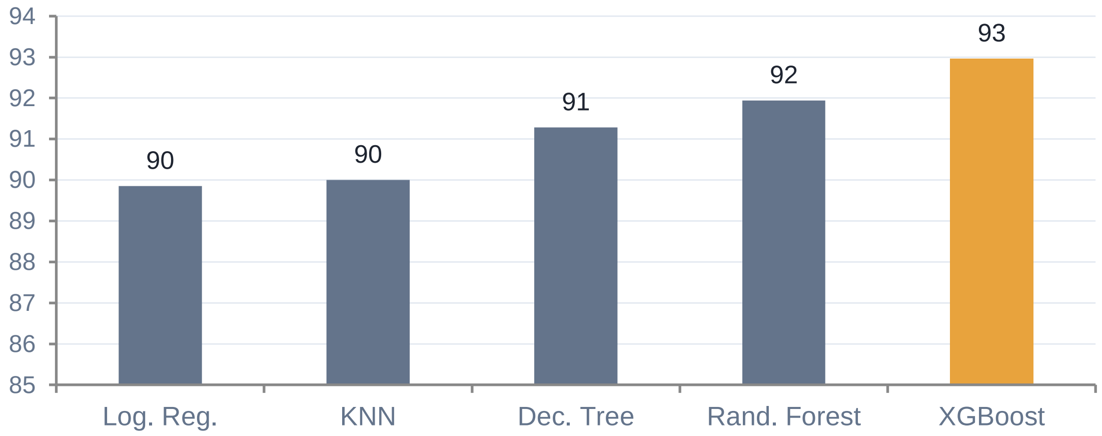
WEEK 3

Model Comparison

Five algorithms evaluated on Accuracy, Precision, Recall, F1-Score, and ROC-AUC

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	85.60%	91.20%	77.36%	83.71%	89.85%
KNN	83.47%	85.21%	79.19%	82.09%	90.00%
Decision Tree	85.28%	90.38%	77.49%	83.44%	91.28%
Random Forest	84.47%	84.77%	82.33%	83.53%	91.94%
XGBoost (Best)	**86.10%**	**91.69%**	**78.01%**	**84.30%**	**92.96%**

ROC-AUC by Model



XGBoost — Best Model

Confusion matrix on 1,597 held-out test records:

779
True Negatives

596
True Positives

54
False Positives

168
False Negatives

WEEK 3

Confusion Matrices — All Five Models

Test set: 1,597 records. Each model classifies unsuccessful (0) vs. successful (1) outcomes

Logistic Regression

776 TN	57 FP
173 FN	591 TP

85.6% correct

KNN

728 TN	105 FP
159 FN	605 TP

83.5% correct

Decision Tree

770 TN	63 FP
172 FN	592 TP

85.3% correct

Random Forest

720 TN	113 FP
135 FN	629 TP

84.5% correct

XGBoost (Best)

779 TN	54 FP
168 FN	596 TP

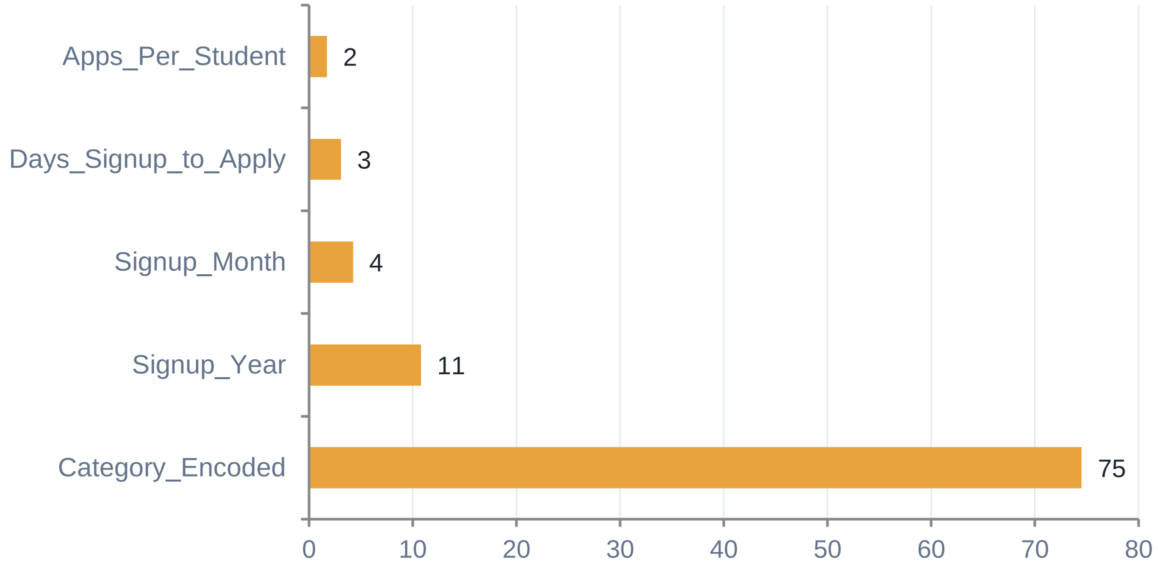
86.1% correct

WEEK 3

What Actually Drives Success?

Feature importance was remarkably consistent across all five algorithms

XGBoost Feature Importance



Cross-Model Comparison

Feature	RF	Tree	XGB	KNN
Category_Encoded	46.1%	85.7%	74.5%	26.4%
Signup_Year	3.4%	3.8%	10.8%	12.2%

Opportunity Category dominates

74.5% of predictive power in XGBoost; 85.7% in the Decision Tree — the single strongest lever.

Timing matters

Signup Year and Signup Month rank 2nd and 3rd, pointing to seasonal, semester-aligned engagement patterns.

Behavior over demographics

Days_Signup_to_Apply and Apps_Per_Student add moderate signal on engagement intent.

No demographic bias detected

Age and Gender consistently rank lowest across all five models.

WEEK 3

Participation Probability & Heatmap Insights

XGBoost probability scores translate into a four-tier priority scale for program managers

Probability → Priority Scale

80% – 100%	High Priority
60% – 79%	Moderate Priority
40% – 59%	Medium Risk
Below 40%	High Risk

Category × Region Heatmap

Africa consistently shows the highest success rates across all opportunity categories, while North America generally records the lowest — regardless of category, geography compounds the internship challenge.

Category × Month Heatmap

Success rates peak during January–February and August–September, aligned with academic semester starts. June–July shows the lowest engagement, likely reflecting summer break.

WEEK 3

Key Drivers & Insights for Program Managers

Model interpretation translated into five actionable operational insights

1	**Focus on Opportunity Design**	Category is the strongest predictor — review structure & requirements of lower-performing categories.
2	**Align Recruitment with Performance Periods**	Schedule campaigns during historically stronger enrollment periods (semester starts).
3	**Use Predictive Scores for Early Intervention**	ROC-AUC of 0.9296 supports reliable identification of at-risk students for targeted outreach.
4	**Monitor Application Timing**	Days_Signup_to_Apply is a consistent predictor — engagement strategies for delayed applicants.
5	**Investigate Regional Differences**	Study why some regions consistently outperform others and replicate successful practices.

WEEK 4

Evidence-Based Insights

Six insights derived from modeling results, prioritized by predictive impact

HIGH

Opportunity Category Design Is the Most Impactful Lever

Courses, Events, and Engagement achieve the highest success; Internships and Competitions consistently underperform.

HIGH

Recruitment Timing Should Align with the Academic Calendar

Success rates peak Jan–Feb and Aug–Sep, coinciding with semester starts; June–July is weakest.

HIGH

Predictive Probability Scores Enable Proactive Intervention

ROC-AUC of 0.9296 supports using the model as an operational early-warning tool, not just research output.

MED

Application Timing Reveals Engagement Levels

Days_Signup_to_Apply ranks 3rd–4th in importance; rapid application suggests higher intrinsic motivation.

MED

Regional Performance Gaps Warrant Investigation

Africa outperforms despite lower volumes; North America underperforms despite higher volumes.

LOW

No Evidence of Demographic Bias

Age and Gender produce the lowest feature importance across all models — an important equity finding.

WEEK 4

Prioritized Actionable Recommendations

Model outputs translated into a strategy with clear rationale for each priority

1

Redesign Lower-Performing Categories

Review structure, commitment demands, and application requirements for Internships and Competitions.

Rationale

Opportunity Category = 74.5% of XGBoost's predictive weight — the single highest-leverage improvement area.

2

Align Recruitment with Academic Calendars

Concentrate marketing and recruitment budgets on Jan–Feb and Aug–Sep windows.

Rationale

Model heatmaps and Signup_Month importance (ranked 3rd) confirm engagement peaks at semester start.

3

Implement Early-Warning Outreach

Deploy XGBoost probability scores to flag students below 60% success probability.

Rationale

Shifts the program from reactive tracking to proactive, targeted mentoring and intervention.

WEEK 4

Scalable Solutions & Measurement Framework

Tactical mechanisms and the KPIs used to track their success

Scalable Solutions

Automated Behavioral Nudges

SMS/email nudge if a student registers but doesn't apply within 14 days.

Behavioral Segmentation

Segment by application velocity and category interest, not demographics.

Cross-Regional Knowledge Transfer

Study Africa's mentorship/peer-referral structures for replication elsewhere.

Measurement Framework (KPIs)

KPI	Metric
Category Conversion Lift	MoM success rate for Internships & Competitions
Application Velocity	Avg. Days_Signup_to_Apply across new cohorts
High-Risk Retention Rate	Actual success of students initially <60% score

Model Reliability & Limitations

Confidence rests on consistent findings across 5 algorithms, ROC-AUC 0.8985–0.9296, and F1-Scores 82–84%. Limitations: historical scope, limited behavioral variables, regional imbalance, no academic-performance data, and a binary success definition.

WEEK 4

Personal Insights & Process Suggestions

Reflections from the internship experience and recommendations for future cohorts

Personal Insights

Professional Communication

Distributed, international teamwork demanded precise, structured updates across time zones.

Cross-Time-Zone Collaboration

Missing real-time task windows underscored the value of proactive check-ins and visible availability.

Self-Awareness on Work Conditions

Clarified what supports peak performance: structured allocation, clear onboarding, overlapping hours.

Process Improvement Suggestions

Structured Task Allocation

Replace informal, real-time task claiming with a structured allocation process at the start of each sprint — e.g. a shared board where roles are assigned transparently regardless of time zone.

Skill Mapping at Onboarding

Document each intern's skill set at the start and share it with the team lead, so assignments intentionally stretch actual capabilities rather than relying on self-advocacy in a fast-moving group.

Participation Success Is Driven by Design and Timing — Not Demographics

86.10%

XGBoost accuracy

0.93

ROC-AUC score

74.5%

importance: Opportunity Category

8,547

records analyzed

From data cleaning through predictive modeling, four weeks of work converged on one clear narrative: redesigning internship pathways, aligning recruitment with the academic calendar, and deploying early-warning outreach offer the platform's highest-leverage opportunities to improve participation outcomes.

Thank you.